FINAL EXAMINATION IN CALCULUS II - SEMESTER: 2022.2

Elitech programs **Total time: 90 minutes**

Attention: No document is allowed.

Prob 1. (1 point) Find the curvature of the curve $y = x^2 - 1$ at $A(1, 0)$.

Prob 2. (1 point) Find the directional derivative of the function $u(x, y, z) = x^2 + 2xy^2 - yz^3$ in the direction of $\overrightarrow{l} = (1, -2, 2)$ at the point $A(1, 1, 1)$.

Prob 3. (1 point) Evaluate $\displaystyle\iint_D (1 + x + y^2)dxdy$, where $D : x^2 + y^2 \leq 1$.

Prob 4. (1 point) Evaluate $\displaystyle\iiint_V (3x + z)dxdydz$, where $V : x^2 + y^2 + z^2 \leq 2z$.

Prob 5. (1 point) Evaluate $\displaystyle\int_0^{+\infty} \frac{\sqrt{x}}{(1 + x^2)^3} dx$.

Prob 6. (1 point) Evaluate $\displaystyle\int_C x + 2y\, ds$, where C: $y = \sqrt{2x - x^2}$.

Prob 7. (1 point) Evaluate $\displaystyle\oint_C (xy + 3x + 2y)\, dx + (xy - 2y)\, dy$, where C is the circle $x^2 + y^2 = 2x$ with counterclockwise orientation.

Prob 8. (1 point) Evaluate $\displaystyle\iint_S (2x - y + z^2)\, dS$, where S is the hemisphere

$$S : x^2 + y^2 + z^2 = 1, x \geq 0.$$

Prob 9. (1 point) Find the flux of the vector field $\overrightarrow{F} = x\vec{i} + y\vec{j} + (z^2 - 1)\vec{k}$ across S, where S is the part of the ellipsoid $x^2 + \dfrac{y^2}{4} + z^2 = 1$, $z \geq 0$, with upward orientation.

Prob 10. (1 point) Find the circulation of the vector field

$$\overrightarrow{F} = (2xze^{x^2} + y^2 - z)\,\overrightarrow{i} + (y - 3z)\,\overrightarrow{j} + (e^{x^2} + x + 2y)\,\overrightarrow{k}$$

around C. Here C is the curve of intersection of the plane $x + y + z = 1$ and the cylinder $x^2 + y^2 = 2y$, oriented counterclockwise as viewed from above.

Chúc các bạn hoàn thành tốt bài thi

LỜI GIẢI CHI TIẾT MÔN GIẢI TÍCH II NHÓM NGÀNH CTTT

Thực hiện bởi team Giải tích II - CLB Hỗ trợ Học tập

Prob 1. ***(1 point)*** *Find the curvature of the curve* $y = x^2 - 1$ *at* $A(1, 0)$.

[Solution]

+) We apply curvature formula of curve $y = f(x)$:

$$C = \frac{|f''(x)|}{\left|1 + [f'(x)]^2\right|^{3/2}}$$

+) We have: $f'(x) = 2x, f''(x) = 2 \quad \Rightarrow f'(1) = 2, f''(1) = 2$

$$\Rightarrow C(A) = \frac{|f''(1)|}{\left|1 + [f'(1)]^2\right|^{3/2}} = \frac{2}{(1 + 2^2)^{3/2}} = \frac{2\sqrt{5}}{25}$$

+) Conclusion: The curvature of the curve at A is $\dfrac{2\sqrt{5}}{25}$

Prob 2. ***(1 point)*** *Find the directional derivative of the function* $u(x, y, z) = x^2 + 2xy^2 - yz^3$ *in the direction of* $\overrightarrow{l} = (1, -2, 2)$ *at the point* $A(1, 1, 1)$.

[Solution]

+) The function $u(x, y, z) = x^2+2xy^2-yz^3 \quad \Rightarrow \overrightarrow{grad u} = (u'_x, u'_y, u'_z) = (2x+2y^2, 4xy-z^3, -3yz^2)$

+) Normalize $\overrightarrow{l}$, we get:

$$\overrightarrow{l'} = \frac{\overrightarrow{l}}{|\overrightarrow{l}|} = (1, -2, 2)\frac{1}{\sqrt{1^2 + (-2)^2 + 2^2}} = \left(\frac{1}{3}, \frac{-2}{3}, \frac{2}{3}\right)$$

+) $\dfrac{\partial u}{\partial \overrightarrow{l}} = \overrightarrow{grad u} \cdot \overrightarrow{l'} = \dfrac{2x + 2y^2}{3} + \dfrac{-2(4xy - z^3)}{3} + \dfrac{2(-3yz^2)}{3}$

$\Rightarrow \dfrac{\partial u}{\partial \overrightarrow{l}}(A) = \dfrac{2 + 2 - 2(4 - 1) + 2(-3)}{3} = \dfrac{-8}{3}$

+) Conclusion: The directional derivative of the function u in the direction of l at the point A is $\dfrac{-8}{3}$

Prob 3. ***(1 point)*** *Evaluate* $\displaystyle\iint_D (1 + x + y^2)dxdy$, *where* $D : x^2 + y^2 \leq 1$.

[Solution]

+) Denoting $I = \displaystyle\iint_D (1 + x + y^2)dxdy$, $D : x^2 + y^2 \leq 1$.

+) Because D symmetry over Oy so $f(x, y) = x$ odd with respect to x.

$$\Rightarrow \iint\limits_D x dx dy = 0 \quad \Rightarrow I = \iint\limits_D (1 + y^2) dx dy$$

+) Define $\begin{cases} x = r\cos(\varphi) \\ y = r\sin(\varphi) \end{cases}$ with $\begin{cases} 0 \le \varphi \le 2\pi \\ 0 \le r \le 1 \end{cases}$

+) Thus, we have:

$$I = \int_0^{2\pi} d\varphi \int_0^1 r(1 + r^2 sin^2(\varphi))dr = \int_0^{2\pi} d\varphi \int_0^1 (r + r^3 sin^2(\varphi))dr$$
$$= \int_0^{2\pi} \left(\frac{1}{2} + \frac{1}{4}\sin^2(\varphi)\right) d\varphi = \int_0^{2\pi} \left(\frac{5}{8} - \frac{1}{8}\cos(2\varphi)\right) d\varphi$$
$$= \frac{5}{8}\varphi\Big|_0^{2\pi} - \frac{1}{16}\sin(2\varphi)\Big|_0^{2\pi} = \frac{5\pi}{4}$$

+) Conclusion: $I = \dfrac{5\pi}{4}$

Prob 4. ***(1 point)*** *Evaluate* $\iiint\limits_V (3x + z) dx dy dz$, *where* $V : x^2 + y^2 + z^2 \le 2z$.

[Solution]

+) Denoting $I = \iiint\limits_V (3x + z) dx dy dz$, where $V : x^2 + y^2 + z^2 \le 2z \Leftrightarrow x^2 + y^2 + (z - 1)^2 \le 1$.

+) Because V symmetry over (Oyz) so $f(x, y, z) = 3x$ odd with respect to x.

$$\Rightarrow \iiint\limits_V 3x dx dy dz = 0. \quad \Rightarrow I = \iiint\limits_V z dx dy dz.$$

+) Define: $\begin{cases} x = r\sin(\theta)\cos(\varphi) \\ y = r\sin(\theta)\sin(\varphi) \\ z = r\cos(\theta) + 1 \end{cases}$ with $\begin{cases} 0 \le \theta \le \pi \\ 0 \le \varphi \le 2\pi \\ 0 \le r \le 1 \end{cases}$

+) Thus, we have:

$$I = \int_0^{\pi} d\theta \int_0^{2\pi} d\varphi \int_0^1 r^2\sin(\theta)(1 + r\cos(\theta))dr = 2\pi \int_0^{\pi} \sin(\theta) d\theta \int_0^1 (r^2 + r^3\cos(\theta))dr$$
$$= 2\pi \int_0^{\pi} (\frac{1}{3}\sin(\theta) + \frac{1}{8}\sin(2\theta))d\theta = 2\pi(-\frac{1}{3}\cos(\theta) - \frac{1}{16}\cos(2\theta))\Big|_0^{\pi}$$
$$= \frac{4\pi}{3}$$

+) Conclusion: $I = \dfrac{4\pi}{3}$

Prob 5. ***(1 point)*** *Evaluate* $\displaystyle\int_0^{+\infty} \frac{\sqrt{x}}{(1+x^2)^3}dx.$

[Solution]

+) Denoting $I = \displaystyle\int_0^{+\infty} \frac{\sqrt{x}}{(1+x^2)^3}dx.$

+) Let $t = x^2 \Rightarrow x = t^{\frac{1}{2}}$ (since $x \in [0, +\infty) \Rightarrow dx = \frac{1}{2}t^{-\frac{1}{2}}dt.$

+) Change of limits:

x	0	$+\infty$
t	0	$+\infty$

+) We have:

$$\begin{aligned}
I &= \int\limits_0^{+\infty} \frac{t^{\frac{1}{4}}}{(1+t)^3} \cdot \frac{1}{2}t^{-\frac{1}{2}}dt = \frac{1}{2}\int\limits_0^{+\infty} \frac{t^{\frac{3}{4}-1}}{(1+t)^{\frac{3}{4}+\frac{9}{4}}}dt = \frac{1}{2} \cdot B\left(\frac{3}{4}, \frac{9}{4}\right) \\
&= \frac{1}{2} \cdot \frac{\frac{9}{4}-1}{\frac{3}{4}+\frac{9}{4}-1} B\left(\frac{3}{4}, \frac{9}{4}-1\right) = \frac{5}{16} \cdot B\left(\frac{3}{4}, \frac{5}{4}\right) \\
&= \frac{5}{16} \cdot \frac{\frac{5}{4}-1}{\frac{3}{4}+\frac{5}{4}-1} B\left(\frac{3}{4}, \frac{5}{4}-1\right) = \frac{5}{64} \cdot B\left(\frac{3}{4}, \frac{1}{4}\right) \\
&= \frac{5}{64} \cdot \frac{\pi}{\sin\frac{\pi}{4}} \\
&= \frac{5\sqrt{2}\pi}{64}
\end{aligned}$$

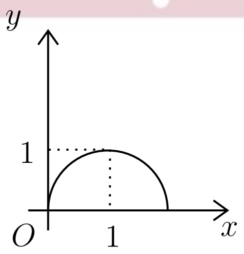
+) Conclusion: $I = \dfrac{5\sqrt{2}\pi}{64}$

Prob 6. ***(1 point)*** *Evaluate* $\displaystyle\int\limits_C x + 2y\, ds$, *where* C: $y = \sqrt{2x - x^2}$.

[Solution]

+) Denoting $I = \displaystyle\int\limits_C x + 2y\, ds$

+) From curve $C : y = \sqrt{2x - x^2} \Rightarrow \begin{cases} (x-1)^2 + y^2 = 1 \\ y \geq 0 \end{cases}$



+) Let $\begin{cases} x = 1 + \cos t \\ y = \sin t \end{cases}$ $(0 \leq t \leq \pi) \Rightarrow \begin{cases} x'_t = -\sin t \\ y'_t = \cos t \end{cases}$

$\Rightarrow ds = \sqrt{(-\sin t)^2 + \cos^2 t}\, dt = dt$

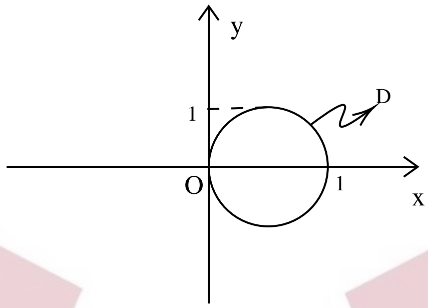
+) We have:

$$I = \int\limits_C x + 2y\, ds = \int_0^{\pi} 1 + \cos t + 2\sin t\, dt = (t + \sin t + 2\cos t)\Big|_0^{\pi} = \pi + 4$$

+) Conclusion: $I = \pi + 4$

Prob 7. ***(1 point)*** *Evaluate* $\oint\limits_C (xy+3x+2y)\,dx+(xy-2y)\,dy$, *where* C *is the circle* $x^2+y^2=2x$ *with counterclockwise orientation.*

[Solution]



+) Let $\begin{cases} P = xy + 3x + 2y \\ Q = xy - 2y \end{cases} \Rightarrow P, Q$ have continuous partial derivatives on $\mathbb{R}^2$

+) We get: $\begin{cases} \dfrac{\partial Q}{\partial x} = y \\ \dfrac{\partial P}{\partial y} = x + 2 \end{cases}$

+) C is closed, bounds region $D : x^2 + y^2 \leq 2x$ and is oriented positively. Applying Green's theorem:

$$I = \iint\limits_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy = \iint\limits_D (y - x - 2)\,dxdy$$

+) Region D is symmetric about $y = 0$ as well as $f(x, y) = y$ is an odd function with respect to y:

$$\Rightarrow \iint\limits_D ydxdy = 0 \Rightarrow I = -\iint\limits_D xdxdy - 2S_D = -\iint\limits_D xdxdy - 2\pi$$

+) Let: $\begin{cases} x = 1 + r\cos\varphi \\ y = r\sin\varphi \end{cases} \Rightarrow |J| = r$ and $D_{(\varphi,r)} \begin{cases} 0 \leq \varphi \leq 2\pi \\ 0 \leq r \leq 1 \end{cases}$

+) We have:

$$\begin{aligned} I &= -\int_0^{2\pi} d\varphi \int_0^1 r(1 + r\cos\varphi)dr - 2\pi \\ &= -\int_0^{2\pi} \left(\frac{\cos\varphi}{3} + \frac{1}{2}\right) d\varphi - 2\pi = -3\pi \end{aligned}$$

+) Conclusion: $I = -3\pi$

Prob 8. ***(1 point)*** *Evaluate* $\iint\limits_S (2x - y + z^2)\,dS$, *where* S *is the hemisphere*

$$S : x^2 + y^2 + z^2 = 1, x \geq 0.$$

[Solution]

+) Denoting $I = \iint\limits_S \left(2x - y + z^2\right) dS$

+) We have $S : x^2 + y^2 + z^2 = 1, x \geq 0 \quad \Rightarrow \begin{cases} x'_y = -\dfrac{y}{\sqrt{1-y^2-z^2}} \\ x'_z = -\dfrac{z}{\sqrt{1-y^2-z^2}} \end{cases}$

$\Rightarrow dS = \sqrt{1 + (x'_y)^2 + (x'_z)^2} dydz = \dfrac{1}{\sqrt{1-y^2-z^2}} dydz$

+) Projection of S on (Oyz) is $D : y^2 + z^2 \leq 1$

$$I = \iint\limits_D \left(2\sqrt{1-y^2-z^2} - y + z^2\right) \cdot \frac{1}{\sqrt{1-y^2-z^2}} dydz$$
$$= \iint\limits_D \left(2 - \frac{y}{\sqrt{1-y^2-z^2}} + \frac{z^2}{\sqrt{1-y^2-z^2}}\right) dydz$$

+) We can see D is symmetric with respect to both y and z

$\Rightarrow \iint\limits_D \dfrac{y}{\sqrt{1-y^2-z^2}} dydz = 0 \quad \Rightarrow I = \iint\limits_D \left(2 - \dfrac{z^2}{\sqrt{1-y^2-z^2}}\right) dydz$

+) Let: $\begin{cases} y = r\cos\varphi \\ z = r\sin\varphi \end{cases} \quad \Rightarrow |J| = r$ and $D_{\varphi,r} \begin{cases} 0 \leq \varphi \leq 2\pi \\ 0 \leq r \leq 1 \end{cases}$

+) We have:

$$I = \int\limits_0^{2\pi} d\varphi \int\limits_0^1 r\left(2 - \frac{r^2\sin^2\varphi}{\sqrt{1-r^2}}\right) dr$$
$$= \int\limits_0^{2\pi} d\varphi \int\limits_0^1 \left(2r + r\sin^2\varphi \cdot \frac{(1-r^2)-1}{\sqrt{1-r^2}}\right) dr$$
$$= \int\limits_0^{2\pi} d\varphi \int\limits_0^1 \left(2r + r\sin^2\varphi\left(\sqrt{1-r^2} - \frac{1}{\sqrt{1-r^2}}\right)\right) dr$$
$$= \int\limits_0^{2\pi} \left(r^2 + \sin^2\varphi \cdot \left(-\frac{1}{3}\sqrt{(1-r^2)^3} + \sqrt{1-r^2}\right)\right)\Big|_0^1 d\varphi$$
$$= \int\limits_0^{2\pi} \left(1 - \frac{2}{3}\sin^2\varphi\right) d\varphi = \int\limits_0^{2\pi} \frac{\cos(2\varphi) + 2}{3} d\varphi$$
$$= \left(\frac{2}{3}\varphi + \frac{1}{6}\sin(2\varphi)\right)\Big|_0^{2\pi} = \frac{4\pi}{3}$$

+) Conclusion: $I = \dfrac{4\pi}{3}$

Prob 9. ***(1 point)*** *Find the flux of the vector field* $\vec{F} = x\vec{i} + y\vec{j} + (z^2 - 1)\vec{k}$ *across* S*, where* S *is the part of the ellipsoid* $x^2 + \dfrac{y^2}{4} + z^2 = 1, z \geqslant 0$*, with upward orientation.*

[Solution]

+) We have $\begin{cases} P(x,y,z) = x \\ Q(x,y,z) = y \\ R(x,y,z) = z^2 - 1 \end{cases}$

+) As we can see from the sketch, S is not a closed region

$\Rightarrow$ We add $S_1 : z = 0$ so we can have $K = S \cup S_1$ is a closed region, with $\begin{cases} x^2 + \dfrac{y^2}{4} + z^2 \leqslant 0 \\ z = 0 \end{cases}$ upward orientation.

$\Rightarrow$ The flux of the vector field we are finding is $I = I_1 - I_2$ with $\begin{cases} I_1 = \oiint\limits_K Pdydz + Qdzdx + Rdxdy \\ I_2 = \iint\limits_{S_1} Pdydz + Qdzdx + Rdxdy \end{cases}$

+) Evaluate I_1:

- Apply Divergence Theorem

$$I_1 = \iint\limits_K F \cdot dS = \iiint\limits_V \text{div}\, F dxdydz = \iiint\limits_V (1 + 1 + 2z) dxdydz$$
$$= \iiint\limits_V (2z + 2) dxdydz$$

- Set $\begin{cases} x = r\cos\varphi\sin\theta \\ y = 2r\sin\varphi \cdot \sin\theta \\ z = r\cos\theta \end{cases} \Rightarrow |J| = 2r^2\sin\theta$ and $V = \begin{cases} 0 \leq r \leq 1 \\ 0 \leq \varphi \leq 2\pi \\ 0 \leq \theta \leq \dfrac{\pi}{2} \end{cases}$

- Thus, we have:

$$I_1 = 2 \cdot \int_0^{2\pi} d\varphi \int_0^{\pi/2} d\theta \int_0^1 2r^2 \sin\theta (1 + r\cos\theta) dr$$
$$= 2 \cdot 2\pi \cdot \int_0^{\pi/2} d\theta \cdot \int_0^1 \left(2r^2\sin\theta + 2r^3\cos\theta\sin\theta\right) dr$$
$$= 4\pi \left(\int_0^{\pi/2} \sin\theta d\theta \cdot \int_0^1 2r^2 dr + \int_0^{\pi/2} \cos\theta\sin\theta d\theta \int_0^1 2r^3 dr \right)$$
$$= 4\pi \left(1 \cdot \frac{2}{3} + \frac{1}{2}\right) = \frac{11\pi}{3}$$

+) Evaluate I_2:

- We can see that $S_1 \begin{cases} z = 0 \\ \vec{n} = (0, 0, -1) \end{cases} \Rightarrow \cos\gamma = \cos(\vec{n}, \overrightarrow{Oz}) < 0$
- We have:

$$I_2 = \iint_D dxdy = \pi \cdot 1 \cdot 2 = 2\pi$$

+) Finally, we have: $I = I_1 - I_2 = \dfrac{11\pi}{3} - 2\pi = \dfrac{5\pi}{3}$

+) Conclusion: the flux of the vector field $\vec{F}$ across S is $\dfrac{5\pi}{3}$

Prob 10. ***(1 point)*** *Find the circulation of the vector field*

$$\overrightarrow{F} = (2xze^{x^2} + y^2 - z)\overrightarrow{i} + (y - 3z)\overrightarrow{j} + (e^{x^2} + x + 2y)\overrightarrow{k}$$

around C. Here C is the curve of intersection of the plane $x+y+z = 1$ and the cylinder $x^2+y^2 = 2y$, oriented counterclockwise as viewed from above.

[Solution]

+) Denote $\begin{cases} P(x,y,z) = 2xze^{x^2} + y^2 - z \\ Q(x,y,z) = y - 3z \\ R(x,y,z) = e^{x^2} + x + 2y \end{cases}$.

+) We have: $\begin{cases} R'_y - Q'_z = 2 - (-3) = 5 \\ P'_z - R'_x = 2xe^{x^2} - 1 - (2xe^{x^2} + 1) = -2 \\ Q'_x - P'_y = 0 - 2y = -2y \end{cases}$

+) Denote S to be the surface whose boundary is C. So S is the plane $x + y + z = 1$.

The orientation of the curve is counterclockwise when viewing from above, thus the unit normal vector of S is $\overrightarrow{n} = \left(\dfrac{1}{\sqrt{3}}; \dfrac{1}{\sqrt{3}}; \dfrac{1}{\sqrt{3}}\right)$, to be suitable with the orientation of the curve C.

+) $P(x,y,z), Q(x,y,z), R(x,y,z)$ are differentiable and continuous functions over the surface S that is bounded by C.

Thus, applying *Stokes* theorem, the circulation of the vector field $\overrightarrow{F}$ is:

$$\begin{aligned} I &= \oint_C Pdx + Qdy + Rdz \\ &= \iint_S (R'_y - Q'_z)dydz + (P'_z - R'_x)dzdx + (Q'_x - P'_y)dxdy \\ &= \iint_S 5dydz - 2dzdx - 2ydxdy \end{aligned}$$

$$= \iint\limits_S 5 \cdot \frac{1}{\sqrt{3}} + (-2) \cdot \frac{1}{\sqrt{3}} + (-2y) \cdot \frac{1}{\sqrt{3}}$$
$$= \iint\limits_S \frac{3-2y}{\sqrt{3}}\, dS$$

+) The surface $S : z = 1 - x - y$ is restricted in the region D which is bounded by $x^2 + y^2 = 2y$. Thus:

$$I = \iint\limits_D \frac{3-2y}{\sqrt{3}} \sqrt{1 + (z'_x)^2 + (z'_y)^2} dxdy$$
$$= \iint\limits_D 3 - 2y\, dxdy$$

+) Let $\begin{cases} x = r\cos\varphi \\ y = 1 + r\sin\varphi \end{cases} \implies \begin{cases} |J| = r \\ 0 \le r \le 1 \\ 0 \le \varphi \le 2\pi \end{cases}$. Thus,

$$I = \int\limits_0^{2\pi} \int\limits_0^1 r(1 - 2r\sin\varphi) dr d\varphi$$
$$= 2\pi \int\limits_0^1 r dr - \int\limits_0^1 2r^2 dr \int\limits_0^{2\pi} \sin\varphi d\varphi$$
$$= \pi.$$

+) Conclusion: The circulation of the vector field around C is π.